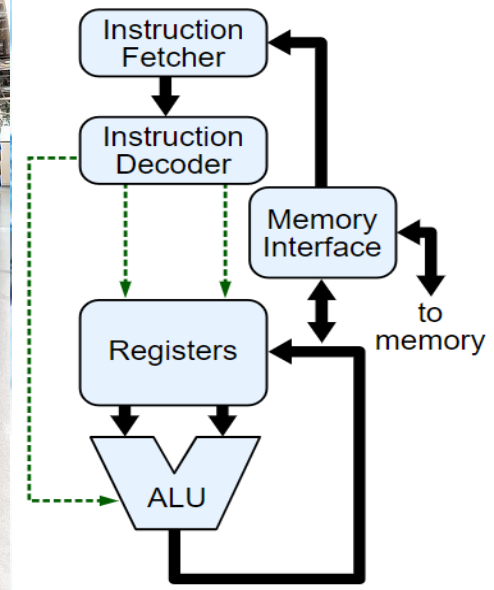
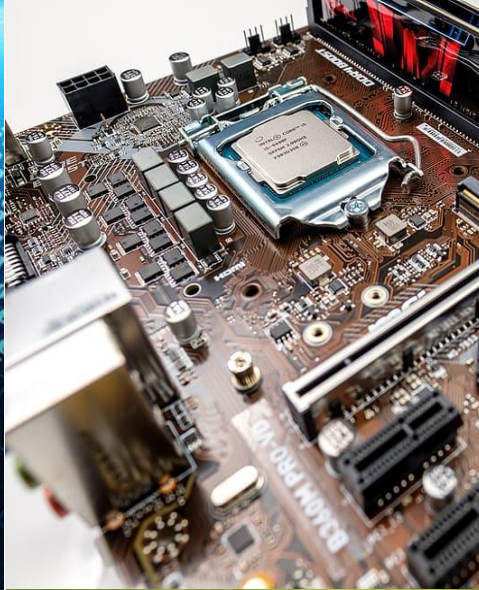




The decimal system

- ▶ 0123456789
 - Ten *digits*.
 - Allows representing large numbers: $193 = 1 \cdot 10^2 + 9 \cdot 10^1 + 3 \cdot 10^0$.
 - A *positional* system.
- ▶ Superior to the older *additive* systems.



1905

UCL's Cruciform Building



Aryabhata
5th century

The decimal system

- ▶ Readily extendable to represent
 - Fractions: $1.7 = 1 \cdot 10^0 + 7 \cdot 10^{-1}$.
 - Rational numbers $7/6 = 1.1\dot{6}$.
 - Aside: Every rational number has a finite repeating representation.
- ▶ Different notations: $\cdot \quad \textcircled{1} \quad \textcircled{2} \quad \textcircled{3} \quad \textcircled{4} \quad \textcircled{5} \quad \textcircled{6} \quad \textcircled{7} \quad \textcircled{8} \quad \textcircled{9}$.
- ▶ Often denoted as 176_{10} .
- ▶ Not the only option!
 - 1,702,131 is the base-1000 number $1 \cdot 1000^2 + 702 \cdot 1000^1 + 131 \cdot 1000^0$.

The binary system

- ▶ Only two digits: 0,1. A digit in $\{0,1\}$ is called binary digit (*bit*).
 - Aside: the *unary* system has a single digit: 1.
- ▶ $1101_2 = 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 = 13_{10}$.
- ▶ Multiplying by 10_2 simply adds a “0” on the right:
 - $1101_2 \cdot 10_2 = 11010_2$.
- ▶ Commonly denoted with the prefix *0b*.
- ▶ Natural for computers!

Binary arithmetic

- ▶ Same as with decimal:

$$\begin{array}{r} 11 \\ 1101_2 = 13 \\ + \\ 0110_2 = 6 \\ \hline 10011_2 = 19 \end{array}$$

$$\begin{array}{r} 1101_2 = 13 \\ \times \\ 0110_2 = 6 \\ \hline 0 \\ 11010 \\ 110100 \\ 000 \\ \hline 1001110_2 = 78 \end{array}$$

...

Converting decimal to binary

► “Greedy algorithm”:

- Find the largest power of two that doesn’t exceed the number.
- Subtract it from the number.
- Repeat.

$$\begin{array}{rclclclclcl} 69_{10} & & 5_{10} & & 1_{10} & & 0_{10} & & \\ & \swarrow & & \swarrow & & \swarrow & & & \\ & + & & + & & + & & & \\ & 64_{10} & & 4_{10} & & 1_{10} & & & \\ & = & & = & & = & & = & \\ & 1000000_2 & & 100_2 & & 1_2 & & = & 1000101_2 \end{array}$$

Hexadecimal representation

▶ Binary numbers are hard (for us puny humans!) to read.

– Contrast 43643_{10} with 1010101001111011_2 .

▶ Base-16 numbers.

– 16 “digits”: 0123456789ABCDEF.

– $0x\text{AA}7\text{B} = 10 \cdot 16^3 + 10 \cdot 16^2 + 7 \cdot 16^1 + 11 \cdot 16^0$.

▶ Easy to convert to binary!

– $1010101001111011_2 \rightarrow$ 

43643_{10}
 $= 0b1010101001111011$
 $= 0x\text{AA}7\text{B}.$

Notation

- ▶ When programming (e.g., in Python/Java) we cannot use subscripts like 17_8 to denote the base.
- ▶ Binary is denoted as `0b*` (e.g., `0b110000011111`).
- ▶ Octal (base 8) is denoted as `0*` (e.g., `06037`).
- ▶ Decimal is without prefix (e.g., `3103`).
- ▶ Hex is denoted as `0x*` (e.g., `0xC1F`).

Test your intuition

- ▶ Which is larger?

Lecture 1: Number systems

Go to

www.menti.com

Enter the code

29 46 96 4



Or use QR code

Test your intuition

► True order:

1st | 010000000 (7 zeros)

2nd | 0x100000 (5 zeros)

3rd | 0b11111111111111111111 (20 ones)

4th | 1000000 (6 zeros)

$$10000000_8 = 8^7 = (2^3)^7 = 2^{21}$$

$$100000_{16} = \underbrace{1000 \dots 0}_2 = 2^{20} = (2^{10})^2 = 1024^2 > 1000^2 = 1000000$$

20 zeros

$$> \underbrace{1111 \dots 1}_2 = 2^{20} - 1 > 1000000$$

20 ones

Representing negative numbers

- ▶ In math, we write -173 for $(-1) \cdot 173$.
 - Similarly, $173 = +173 = (+1) \cdot 173$.
- ▶ This is called a sign-magnitude representation.
- ▶ Not ideal.
 - Zero has two representations $(+0, -0)$.
 - Complicates the calculations.

Representing negative numbers

Two's complement.

- “The two's complement of an N-bit number is defined as its complement with respect to 2^N ”.

$$\begin{array}{rcl} 2^3 & & 3 \\ = & & = \\ 1000_2 - 011_2 & = & 101 \quad (\text{represents } -3) \end{array}$$

$$x = (x_{N-1} x_{N-2} \dots x_0)_2 = \left(-x_{N-1} 2^{N-1} + \sum_{i=0}^{N-2} x_i 2^i \right)$$

Decimal	Two's comp.
-4	100
-3	101
-2	110
-1	111
0	000
1	001
2	010
3	011

Exercise: what 2's comp.

number is represented by 10101?

$$-2^4 + 2^2 + 2^0 = -11$$

Exercise II: what 2's comp.

number is represented by 110101?

$$-2^5 + 2^4 + 2^2 + 2^0 = -11$$

Two's-complement

- ▶ Why should we use 2's-complement?
 - Single zero representation simplifies equality checking.
 - Arithmetic is easier!

$$\begin{array}{r} 0011 = 3 \\ + 1001 = -7 \\ \hline 1100 = -4 \end{array}$$

$$\begin{array}{r} 0110 = 6 \\ + 1101 = -3 \\ \hline 10011 \\ \underbrace{}_{= 3} \end{array}$$

$$\begin{array}{r} \times 1110 = -2 \\ 0011 = +3 \\ \hline 1110 \\ 11100 \\ \hline 101010 \\ \underbrace{}_{= -6} \end{array}$$

Completely transparent! *

What does the following represent?

► `0b0100100001101001`

–In decimal: $2^{14} + 2^{11} + 2^6 + 2^5 + 2^3 + 2^0 = 18537_{10}$.

–In Hex: `0x4869`.

–In ASCII: `'Hi'`.